

```
//Experiment 4: Interfacing IR or Ultrasonic Sensor for Distance  
Measurement and Display on LCD
```

```
#include <Wire.h>  
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);  
const int trigPin = 4;  
const int echoPin = 6;
```

```
void setup() {  
    pinMode(trigPin, OUTPUT);  
    pinMode(echoPin, INPUT);  
  
    Serial.begin(9600);  
  
    lcd.init();  
    lcd.backlight();  
}
```

```
void loop() {  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);
```

```
long duration = pulseIn(echoPin, HIGH, 30000);

lcd.clear();

if (duration == 0) {
    Serial.println("No echo detected (timeout)");
    lcd.setCursor(0, 0);
    lcd.print("No echo detected");
    lcd.setCursor(0, 1);
    lcd.print("(timeout)");
}
else
{
    float distance = duration * 0.0343 / 2.0;
    Serial.print("Distance in cm: ");
    Serial.println(distance, 2);

    lcd.setCursor(0, 0);
    lcd.print("Distance: ");
    lcd.print(distance, 1);
    lcd.print("CM");
    lcd.setCursor(0, 1);
    lcd.print("HC-SR04 Sensor");
}
```

```
    delay(500);  
}
```